

MMALS SplitCIFAR10 RC2O-v2.2c Evidence Analysis

Seeds=3, RUN_MODE=evidence, feature representation=small_cnn_feature/A1 sanity backbone reuse. Analysis generated from the uploaded evidence ZIP.

Verdict: this run is a major positive step. The representation ladder passes, the selected RC2O policy beats validation-selected replay by a large margin, class-collapse is gone on the selected policy, and the specialization audit is now materialized and aligned to the selected candidate. This supports a **SplitCIFAR10 evidence claim**, not CIFAR10 SOTA and not yet broad external-dataset robustness.

1. Key results

Item	Final avg acc	Std	Min-task	Forgetting	Min-class
Selected RC2O	0.909	0.012	0.828	0.026	0.792
Safe readout candidate	0.910				
Best replay baseline	0.741			0.188	
Joint MLP control	0.753				
ZF oracle probe	0.921				

Selected RC2O minus best replay: **0.167**. Safe readout minus replay: **0.168**. ZF oracle minus selected RC2O: **0.012**.

2. Representation ladder

Backbone	Sup val	Sup test	Probe val	Probe test	Params	Trainable
smallcnn_v21_reference	0.750	0.749	0.751	0.745	129290	129290
smallcnn_a1_aug_norm	0.881	0.879	0.882	0.881	625482	625482
resnet18_imagenet_frozen	0.868	0.860	0.868	0.860	11181642	5130

Interpretation: A1 reaches 0.881 supervised validation and 0.882 frozen linear-probe validation. The CL comparison is no longer representation-failure dominated. ResNet18 frozen also passes its probe floor.

3. Leaderboard and baselines

Method	Role	Avg	Std	Min-task	Forget	Min-class
paired_context_gap_safe_class_balanced_readout_calibrated	deployable_mmals_paired	0.910	0.013	0.831	0.025	0.796
paired_context_gap_selected	deployable_mmals_paired	0.909	0.011	0.836	0.023	0.815
paired_context_temp_blend_selected_head_guard_035	deployable_mmals_paired	0.909	0.011	0.836	0.023	0.815
paired_rc2o_v1_feature_only_regime_hard_selector_policy	deployable_mmals_paired	0.909	0.012	0.828	0.026	0.792
paired_proto_global_head_ce_kl_guard_035	deployable_mmals_paired	0.907	0.014	0.831	0.023	0.787
paired_generic_weak_boundary_latent_calibrated	deployable_mmals_paired	0.895	0.036	0.762	0.041	0.733
paired_context_blend_selected_global	deployable_mmals_paired	0.895	0.036	0.762	0.041	0.733
paired_proto_global_no_repair	deployable_mmals_paired	0.894	0.036	0.762	0.039	0.731
paired_context_temp_blend_selected_global	deployable_mmals_paired	0.889	0.047	0.782	0.047	0.754
baseline_joint_upper_bound	diagnostic_upper_bound	0.753	0.014	0.608	0.000	0.540
baseline_experience_replay	deployable_continual_baseline	0.741	0.008	0.601	0.188	0.540
baseline_balanced_replay	deployable_continual_baseline	0.741	0.008	0.601	0.188	0.540

Caution: the joint_upper_bound row should now be described as a joint MLP control, not a theoretical upper bound over all candidate/readout families, because the selected MMALS policy exceeds it.

4. SplitCIFAR10 external-bridge gate

Check	Value	Floor	Ceiling	Pass
a1_supervised_backbone_pass	0.881	0.650		True
a1_frozen_linear_probe_pass	0.882	0.550		True
joint_upper_bound_representation_floor	0.753	0.550		True
selected_rc2o_competitive_with_replay	0.909	0.731		True
selected_min_task_nonzero	0.828	0.050		True
selected_forgetting_not_explosive	0.026		0.200	True
selected_min_class_nonzero	0.792	0.050		True
selected_zero_recall_classes_gate			0.000	True
specialization_components_materialized	4.000	4.000		True
specialization_aligned_to_selected_policy	1.000	1.000		True

All ten gate checks pass in the uploaded evidence run, including A1 sanity, joint control floor, selected-vs-replay, min-task, forgetting, min-class, zero-recall, materialized specialization components, and selected-policy alignment.

5. Readout collapse and class safety

Method	Eff pred classes	Min recall	Collapse rate	Zero-recall by seed
paired_rc2o_v1_feature_only_regime_hard_selector_policy	9.983	0.792	0.000	☐; ☐; ☐
paired_context_gap_safe_class_balanced_readout_calibrated	9.987	0.796	0.000	☐; ☐; ☐
paired_context_gap_selected	9.988	0.815	0.000	☐; ☐; ☐
baseline_balanced_replay	9.808	0.540	0.000	☐; ☐; ☐
baseline_experience_replay	9.808	0.540	0.000	☐; ☐; ☐
baseline_joint_upper_bound	9.888	0.540	0.000	☐; ☐; ☐

Selected RC2O has no zero-recall classes and high effective predicted-class count. This is the main operational difference from earlier failed SplitCIFAR10 smokes.

6. Specialization audit

Component	Rows	Seeds	All OK
host_ablation	90	3	True
host_probe	75	3	True
prototype_separation	30	3	True
route_context_and_confusion	30	3	True

All four required specialization components are present for all three seeds. Host-probe and host-ablation no longer fail, and the status table marks them aligned to the selected candidate. This supports an auditability claim for host specialization. It is still diagnostic unless converted into a causal ablation result section with effect-size plots.

7. Energy probe and LSA diagnostics

Seed	Oracle hit	Success	Wrong ctx	Diverge	Med gap	Energy drop
0.000	1.000	1.000	0.000	0.000	0.295	16.001
1.000	1.000	1.000	0.000	0.000	0.374	14.369
2.000	1.000	1.000	0.000	0.000	0.590	17.540

LSA mean cosine: **0.904**; minimum task/host mean cosine: **0.661**; mean suspicious rate: **0.334**. LSA remains diagnostic-only.

8. Baseline hardening trace

Seed	Baseline	Candidate	Config	Val score	Val avg	Val min-task	Val forgetting
0	finetune	finetune_cand_0	{"LR": 0.001}	0.006	0.178	0.000	0.685
0	ewc	ewc_cand_0	{"EWC_LAMBDA": 1.0}	0.007	0.178	0.000	0.686
0	lwf	lwf_cand_2	{"DISTILL_TEMPERATURE": 3.0, "LWF_LAMBDA": 2.0}	0.221	0.221	0.007	0.003
0	experience_replay	experience_replay_cand_2	{"BASELINE_REPLAY_WEIGHT": 1.0, "MEMORY_PER_CLASS": 192, "MEMORY_PER_TASK": 192}	0.740	0.710	0.565	0.109
0	balanced_replay	balanced_replay_cand_2	{"BASELINE_REPLAY_WEIGHT": 1.0, "MEMORY_PER_CLASS": 192, "MEMORY_PER_TASK": 192}	0.740	0.710	0.565	0.109
0	sparse_moe	sparse_moe_cand_0	{"SPARSE_MOE_ENTROPY_WEIGHT": 0.0}	0.007	0.180	0.000	0.689
0	dense_moe	dense_moe_cand_0	{"LR": 0.001}	0.007	0.180	0.000	0.689
0	hard_top1_moe	hard_top1_moe_cand_0	{"LR": 0.001}	0.011	0.177	0.000	0.667
0	noisy_topk_moe	noisy_topk_moe_cand_1	{"SPARSE_MOE_ENTROPY_WEIGHT": 0.001}	0.007	0.178	0.000	0.686
0	pnn	pnn_cand_1	{"LR": 0.0005}	0.007	0.179	0.000	0.687
0	joint_upper_bound	joint_upper_bound_sanity	{"LR": 0.001}				
1	finetune	finetune_cand_0	{"LR": 0.001}	0.005	0.177	0.000	0.689
1	ewc	ewc_cand_2	{"EWC_LAMBDA": 50.0}	0.005	0.177	0.000	0.686
1	lwf	lwf_cand_2	{"DISTILL_TEMPERATURE": 3.0, "LWF_LAMBDA": 2.0}	0.197	0.197	0.005	0.005
1	experience_replay	experience_replay_cand_1	{"BASELINE_REPLAY_WEIGHT": 1.0, "MEMORY_PER_CLASS": 96, "MEMORY_PER_TASK": 96}	0.723	0.700	0.555	0.130
1	balanced_replay	balanced_replay_cand_1	{"BASELINE_REPLAY_WEIGHT": 1.0, "MEMORY_PER_CLASS": 96, "MEMORY_PER_TASK": 96}	0.723	0.700	0.555	0.130
1	sparse_moe	sparse_moe_cand_1	{"SPARSE_MOE_ENTROPY_WEIGHT": 0.001}	0.009	0.178	0.000	0.676
1	dense_moe	dense_moe_cand_0	{"LR": 0.001}	0.006	0.177	0.000	0.683

The baseline-hardening artifact has 72 candidate rows (3 seeds). It shows validation-selected replay memory sizes, LwF lambda/temperature, EWC lambda, MoE weights, PNN LR, and the joint MLP control. This addresses the arbitrary-default baseline criticism for this run.

9. Scientific interpretation and next step

Claim now supported: SplitCIFAR10 3-seed evidence that RC2O/MMALS with A1 features outperforms validation-selected replay under this protocol, with lower forgetting and no class collapse.

Claim still not supported: SOTA CIFAR10 performance, broad external visual robustness, or positive backward transfer.

Methodological correction: rename `joint_upper_bound` to `joint_mlp_control` or `joint_same-readout-control` in the paper; it is not a theoretical upper bound once MMALS candidates exceed it.

Next experiment: run a 5-seed robust SplitCIFAR10 confirmation or move to SplitCIFAR100 only after freezing this exact A1 reuse, gate, and baseline-hardening protocol.